Homework 7

(Due time:21:30,Apr.30,2026)

Calculate $\dot{\theta}_{12}$, $\dot{\theta}_{23}$, t_1, t_2, and t_3 for a two-segment linear function with parabolic blends (LFPB). For this joint, $\theta_1 = 5.0°$, $\theta_2 = 15.0°$, $\theta_3 = -10.0\ ◦$. Assume that $t_{d12} = t_{d23} = 1.0$ seconds and that the default acceleration to use during blends is 80 degrees/second^2. Sketch plots of position, velocity, and acceleration of θ.

$$\dot{\theta}_{12} = \frac{\theta_2 - \theta_1}{t_{d12}} = \frac{15° - 5°}{1\,s} = 10°/s$$

$$\dot{\theta}_{23} = \frac{\theta_3 - \theta_2}{t_{d23}} = \frac{-10° - 15°}{1} = -25°/s$$

$$t_1 = \frac{|\dot{\theta}_{12} - 0|}{|\ddot{\theta}|} = \frac{10}{80} = 0.125$$

$$t_2 = \frac{|\dot{\theta}_{23} - \dot{\theta}_{12}|}{|\ddot{\theta}|} = \frac{|-25 - 10|}{|-80|} = 0.4875$$

$$t_3 = \frac{|0 - \dot{\theta}_{23}|}{|\ddot{\theta}|} = \frac{|0 - (-25)|}{80} = 0.3125$$

